



FlowForge

Autonomous Exception Resolution for the Just-In-Time Economy

FAR AWAY 2026 — Finale · Japan

Team: {{team names}}

KI · 起 — CONTEXT

Japan's Quiet Crisis

The workforce that keeps supply chains running is shrinking. AI agents are no longer optional — they are national infrastructure.

15M

Working-Age Decline

Population lost over 20 years — and the trend continues

40%

AI-Embedded Apps

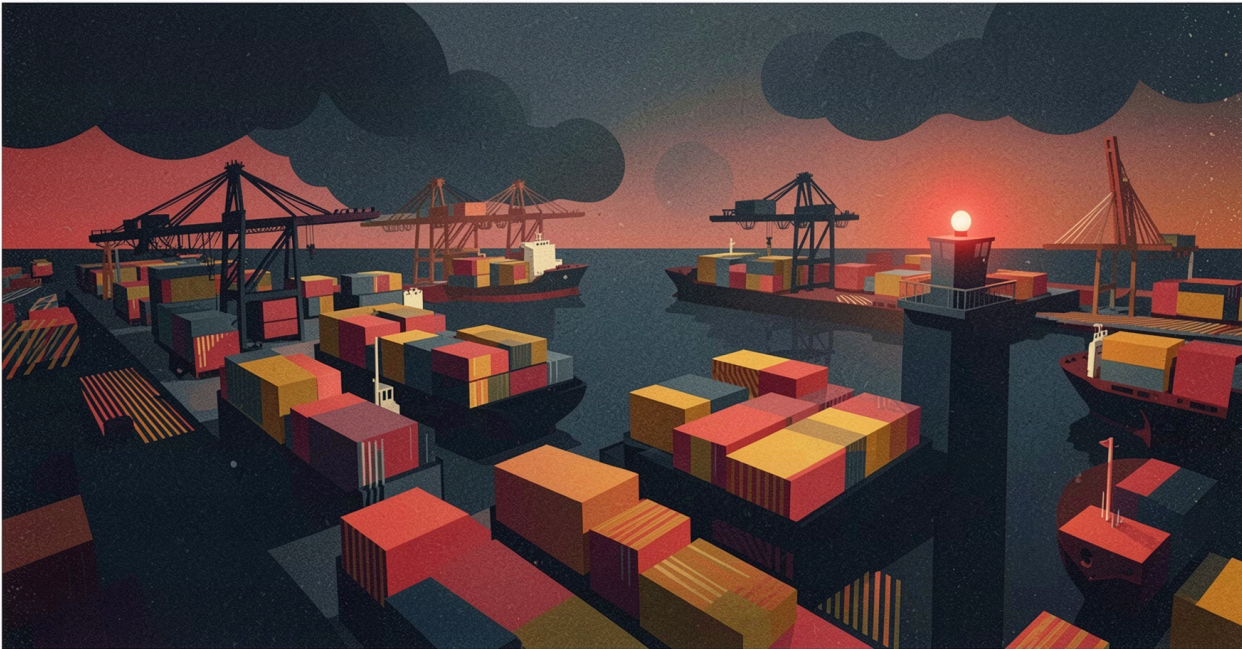
Enterprise applications projected to embed AI agents by 2026

5.0

Society 5.0 / 社会5.0

METI's national strategy: agentic AI as public infrastructure

When Automation Breaks



Deterministic automation handles the **routine 80%** — purchase orders, scheduled shipments, standard workflows. But gray-area exceptions shatter it completely:

- **Port closures** from weather or geopolitical events
- **Supplier failures** with no pre-scripted contingency
- **Regulatory disruptions** requiring real-time judgment

These demand human firefighting — exactly the capacity Japan is running out of. The gap between routine and exception is where resilience collapses.

A Team of Agents, Not a Wrapper

FlowForge is a closed-loop agent team. Each stage reasons, plans, and verifies — escalating to humans only when judgment is irreplaceable.

Watcher

Monitors real-time signals across ports and suppliers

Audit Trail

Logs every decision for full traceability

Executor

Acts autonomously on verified plans



Diagnosis

Identifies root cause and classifies exception severity

Planner

Generates candidate resolution strategies

Verifier

Red-teams every plan before execution

Human Gate

Escalates only when judgment is needed

Reasoning Meets Rigor

Four specialized layers — each with a distinct role — ensure FlowForge is trustworthy, not just clever. The LLM reasons; the solver computes; the verifier red-teams; the simulator stress-tests.

The diagram consists of four horizontal arrows pointing to the right, stacked vertically. Each arrow represents a layer in the FlowForge architecture. The arrows are colored in a gradient from dark brown at the top to light orange at the bottom. The text for each layer is written in bold black font inside the arrow.

LLM Strategist

OR-Tools Solver

Verifier Agent

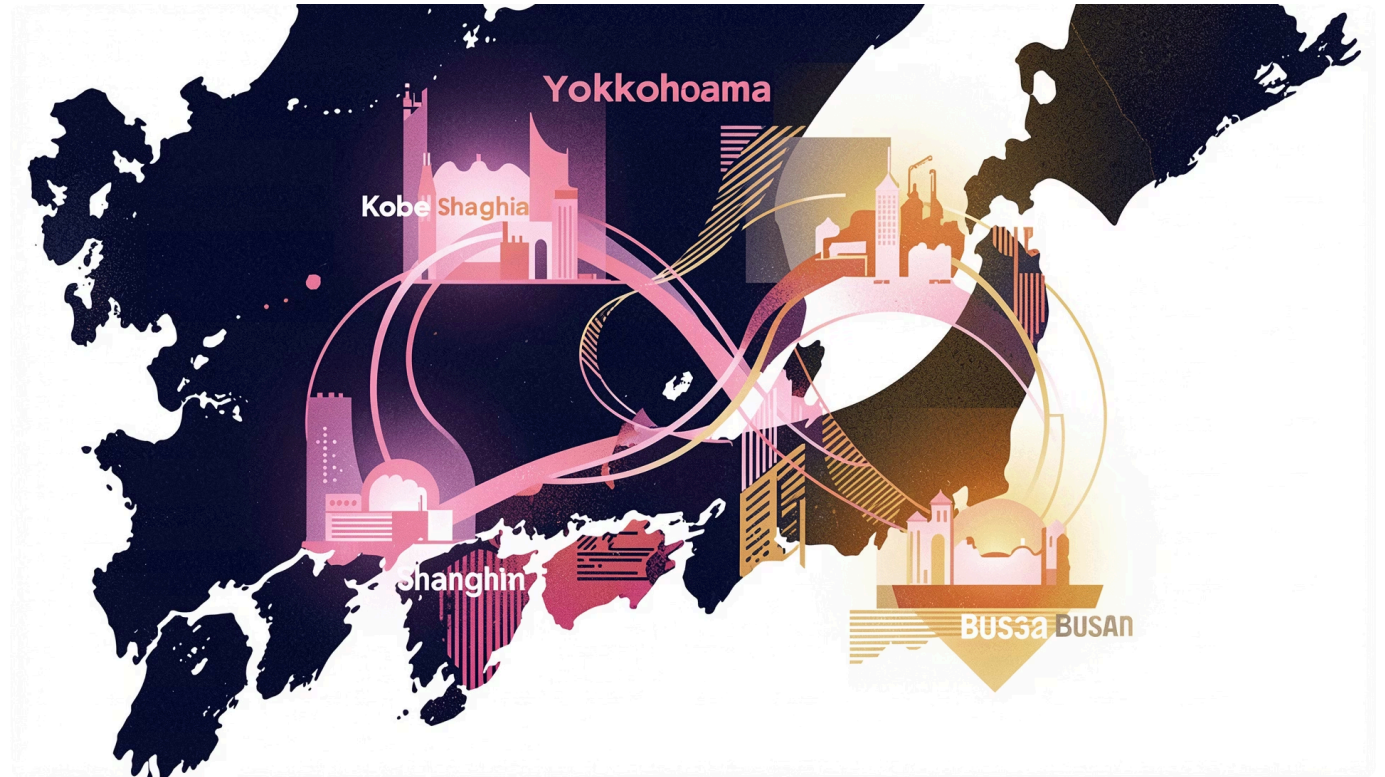
Monte Carlo

This is why FlowForge is **trustworthy** — not just clever. Every action is reasoned, computed, challenged, and stress-tested before execution.

Live, and Measured

FlowForge monitors **four real ports** — Yokohama, Kobe, Shanghai, Busan — with live weather and disruption feeds. One labeled simulated disruption is used for reproducibility, fully transparent.

i Human workload remains flat as exception volume scales — the core promise of agentic AI.



{{X}}% Auto-Resolved

Exceptions handled without human intervention

{{Y}}% Escalated

Cases requiring human judgment — only when necessary

\${Z} Protected

Business value preserved through autonomous resolution



KETSU · 結 — IMPACT

One Engine, Many Domains

The same agent team extends from logistics to manufacturing, from Japan to any country — a config change, not a rewrite.



Manufacturing

Production line disruptions,
component shortages, quality
escalations



Geographic Scale

Japan-first, globally portable —
regional compliance via
configuration



Workforce Resilience

Operations stay robust as the labor pool contracts — built for a
society with fewer people to spare